ANALYTICS ENGINEERING · PORTFOLIO PROJECT

ShopFlow Analytics

How I turned messy e-commerce data into answers for the CFO, CMO and Ops head.

Python | SQL | dbt | DuckDB | Streamlit | Plotly | GitHub Actions

~41k
orders modelled

10,000
customers

28
dbt models

140+
automated data tests

Swipe → 6 business insights inside

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THE PIPELINE

Raw data in. Business-ready marts out.

Messy source data becomes tested, business-ready marts in 7 steps.

1. **Source data**
   orders · items · payments · customers · products · sellers · reviews · marketing spend
2. **Extract & load**
   raw CSVs landed in DuckDB, untouched
3. **Staging · 8 views**
   type, clean, dedupe, standardise labels
4. **Intermediate · 6 views**
   roll-ups, order sequencing, RFM scores
5. **Core star schema**
   fct_orders · fct_order_items · dim_customers · dim_products · dim_date
6. **Marts**
   finance · marketing · operations
7. **Streamlit dashboard**
   4 tabs for leadership, reads marts only

Quality gate: GitHub Actions runs the full dbt build and 140+ tests on every push.

INSIGHT 1 · FINANCE

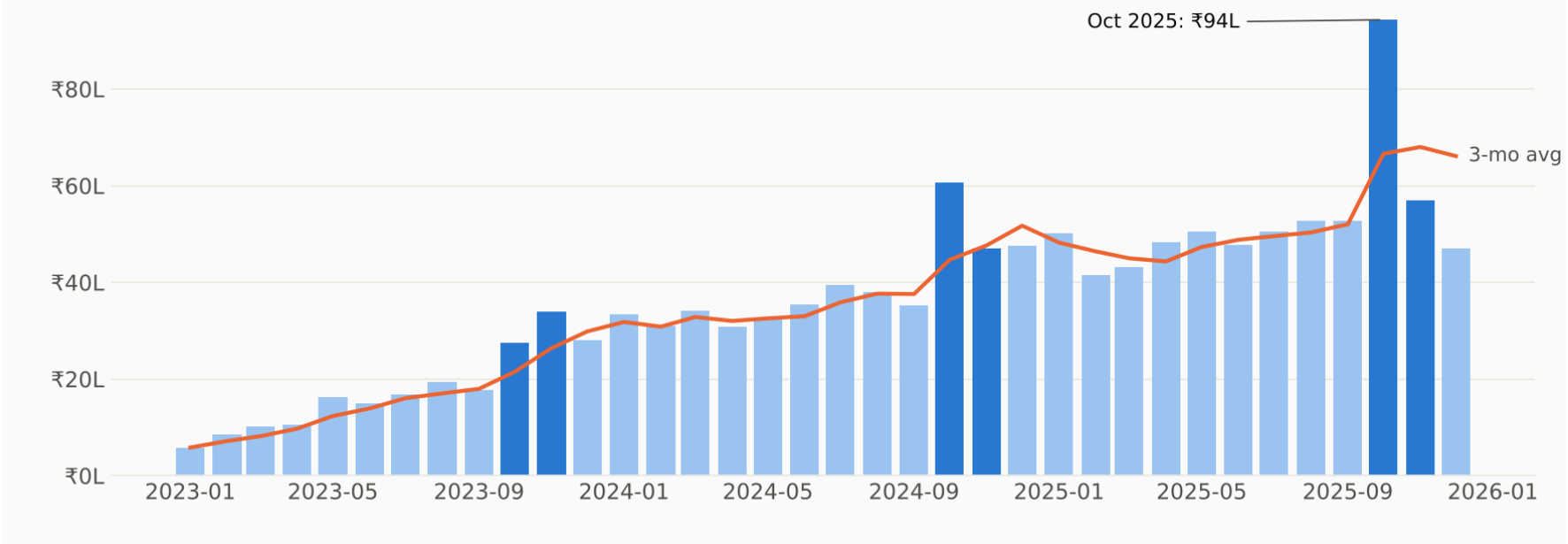
+37%

Revenue growth, like-for-like

Apr–Dec 2025 net revenue was ₹5.01 Cr vs ₹3.67 Cr for the same months of 2024, with gross margin steady at ~25%.

Net revenue by month

Dark bars = festive season (Oct–Nov). Revenue recognised on delivered orders only.



₹5.01Cr
Apr–Dec 2025 net revenue

₹94L
peak month (Oct 2025)

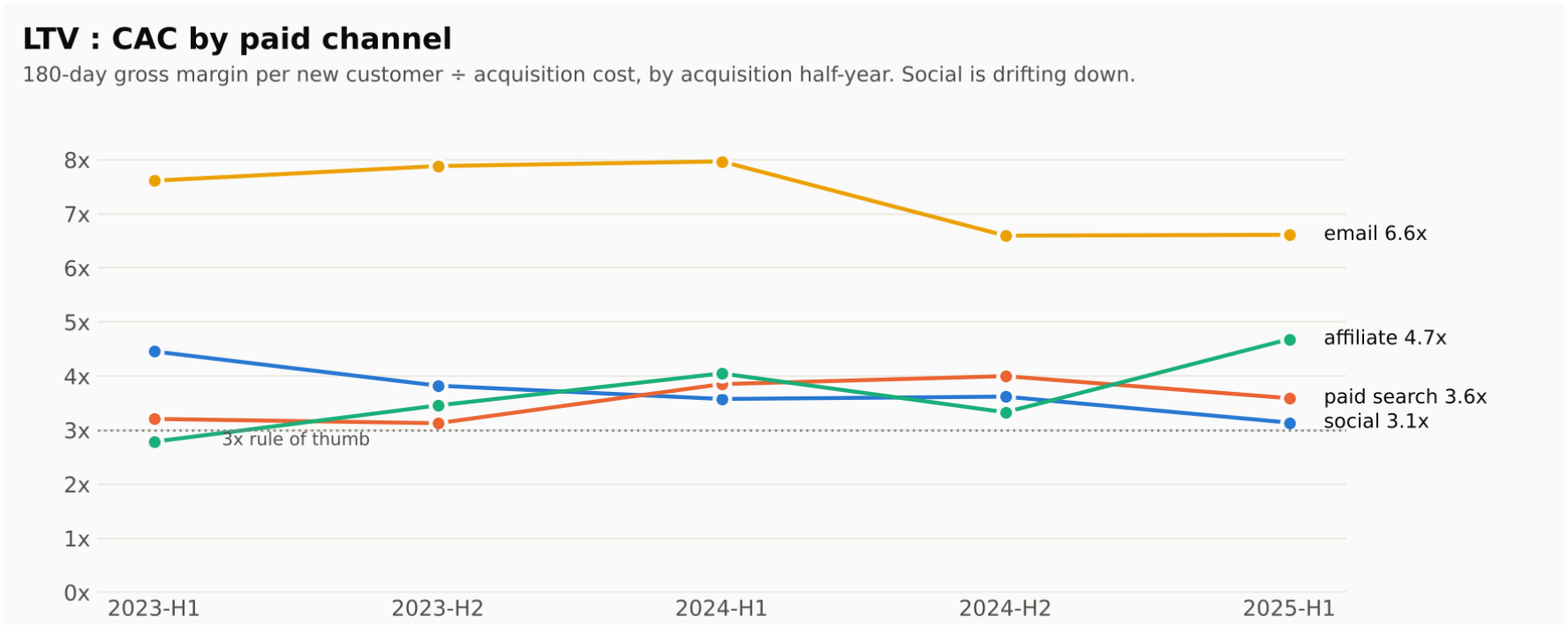
~25%
gross margin, held steady

So what: Festive Oct–Nov spikes inflate totals, so I compare the same months year-on-year.

INSIGHT 2 · MARKETING

Social media is getting expensive. Affiliate is winning.

180-day LTV:CAC for social fell from 4.4x to 3.1x, close to the 3x floor. Affiliate climbed to 4.7x.



3.1x
social (was 4.4x)

4.7x
affiliate, rising

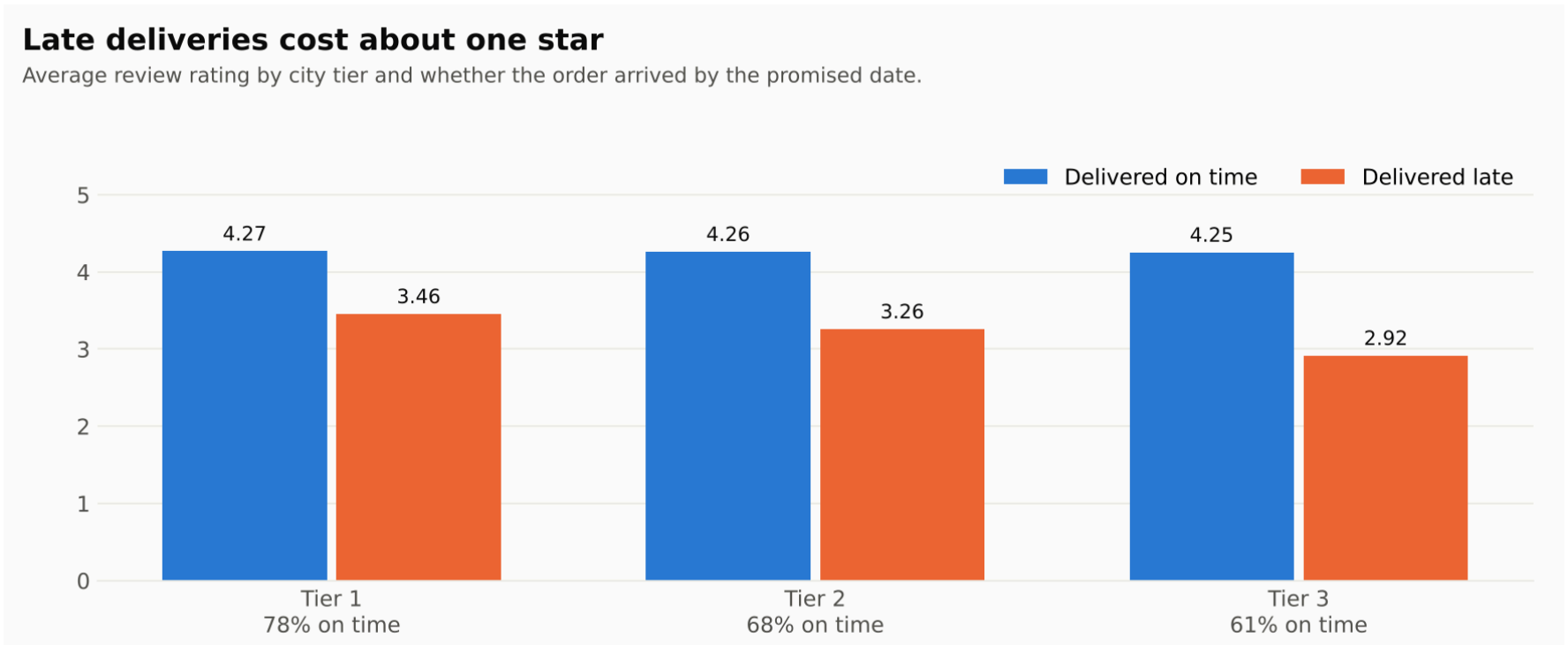
6.6x
email, best channel

Recommendation: Shift incremental budget from social to affiliate and paid search, and re-test social creatives.

INSIGHT 3 · OPERATIONS

A late delivery costs about one star

On-time orders average ~4.3★ in every city tier. Late orders drop to 2.9★ in Tier 3, where only 61% of orders arrive on time.



78%
on time · Tier 1

68%
on time · Tier 2

61%
on time · Tier 3

So what: Fixing Tier 3 logistics is a customer-experience lever, not just a cost question.

INSIGHT 4 · CUSTOMERS (RFM)

62%

41% of customers bring in 62% of revenue

RFM scoring splits customers into 8 segments, each with a recommended CRM action.

3,438

Champions + Loyal customers

704

“Cannot Lose Them”: high-value customers who went quiet

₹2.6 Cr

historical revenue held by that at-risk group

~1 in 3

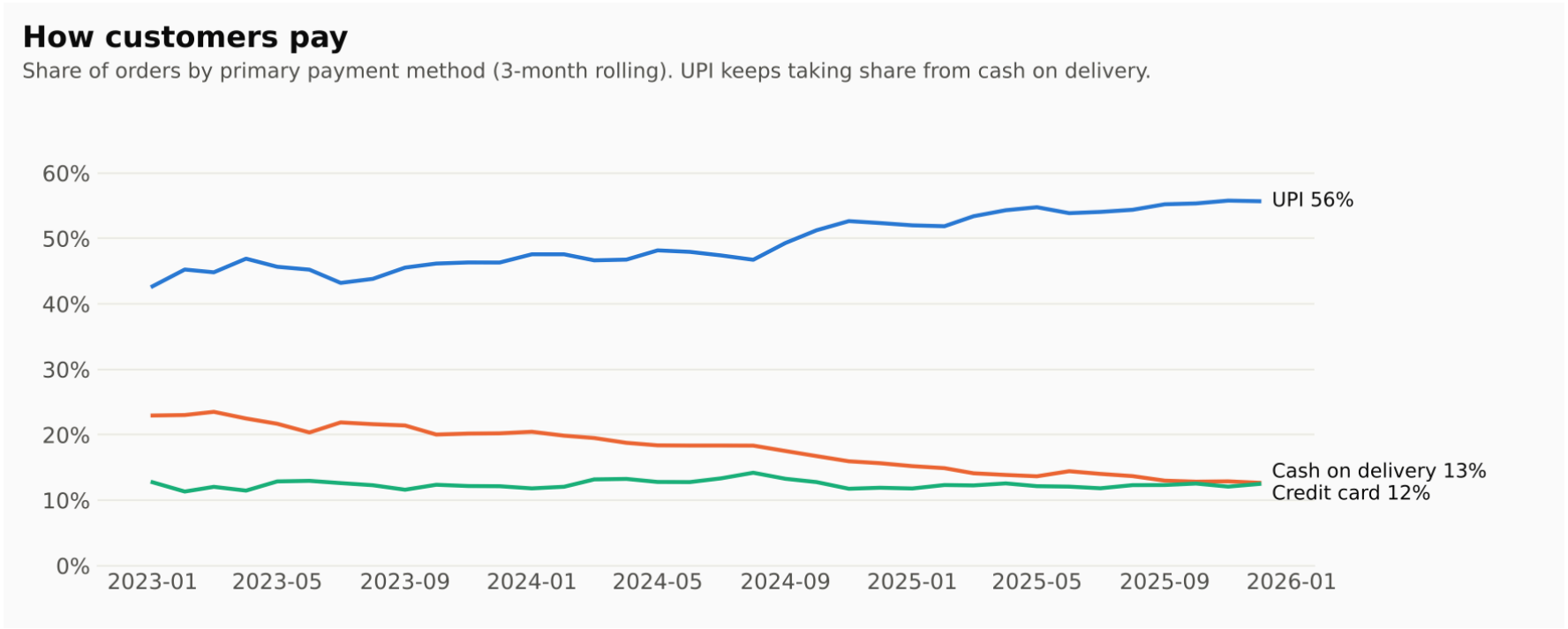
new customers order again in month 1 (cohort analysis)

Top priority: a win-back campaign for the 704 “Cannot Lose Them” customers.

INSIGHT 5 · PAYMENTS

UPI keeps replacing cash on delivery

UPI's share of orders rose from 43% to 56%, while cash on delivery fell from 23% to 13%.



56%

UPI share (from 43%)

13%

cash on delivery (from 23%)

12%

credit card, flat

So what: Fewer failed COD deliveries and faster cash collection.

UNDER THE HOOD

Built like a production data stack

- **Layered dbt modelling** staging → intermediate → star schema → marts
- **140+ data tests** incl. finance reconciliation: KPIs tie to the fact table to the rupee
- **Unit tests** for order sequencing and every RFM segment rule
- **Messy-data handling** CDC duplicates, 15+ spellings of payment methods
- **Incremental models + SCD Type 2** price history via dbt snapshots
- **Indian fiscal calendar** Apr–Mar years and festive sale calendar as macros
- **CI/CD with GitHub Actions** build, test and publish docs on every push

EXPLORE IT YOURSELF

Try the live dashboard

4 tabs: Executive overview · Customers & retention · Marketing efficiency · Operations

LIVE DASHBOARD
shopflow-analytics-nh2a9ucu75xyequqmv94cc.streamlit.app

GITHUB REPOSITORY
github.com/Dharmeshkumar2327/shopflow-analytics

Note: built on a synthetic, seeded dataset of a fictional Indian marketplace. The numbers show the method, not a real company's results.

Feedback welcome. Which insight would you act on first?